



The Future of

The battleground of the future will not be soaked with human blood—intelligent machines will do the fighting, and the killing

The cutting edge of technology has affected the equations of a war so dramatically that the motivations and conventions of a typical 'war' have become vastly different from what they were earlier. Despite this apparent shift, the fundamental nature of war does not change with technology, and understanding how war does not change is key to understanding the art of war itself. In this article, we take the ancient Chinese wisdom of Shang Tsu, and draw parallels with the present day scenario of laser-guided bombs, night vision, mechanised warriors, and killer warplanes.

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Knowledge of the enemy's dispositions can only be obtained from other men

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The art of war

This statement needs some revision, especially in the wake of spy satellites, reconnaissance planes, and surveillance systems. Today's intelligence gathering is much more passive in nature, and doesn't involve using spies that much. Take the example of Echelon. Several credible reports suggest that this project attempts to capture staggering volumes of satellite, microwave, cellular, and fibre-optic traffic, including communications. This vast quantity of voice and data communications is then processed through sophisticated filtering technologies. Most of what is known—or believed to be known—about Echelon is based on rumours and allegations, such as the claims that the system intercepts 3 billion e-mails every day, or sorts through 90 per cent of the traffic on the Internet. Echelon's computers are supposedly capable of recognising key words, names, telephone numbers, and voices. They

gather information from listening posts all around the world, with additional input from about 120 US satellites.

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The skillful soldier does not raise a second levy, neither are his supply-wagons loaded more than twice

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The art of war

The high-tech foot soldier of the near future will bear a closer resemblance to the Terminator. That is the promise of the Land Warrior, a 9-pound 'back-top' computer that manages a global positioning receiver, miniature camera, laser rangefinder, video and thermal weapons sights, and a radio. The 21st Century Land Warrior (21CLW) is the name given to the concept of new future military development for the man on the ground. This concept has been a study and a progressive objective for the past decade for many Western military forces. The weapon subsystem is built around the M-16/M-4 modular carbine. It has a laser range finder/digital compass, a daylight video camera, a laser aiming light, and a thermal sight.

Cables run from the battery-powered backpack to a soldier's helmet or sunglasses, where a tiny 'heads-up' display monitor is mounted, and his weapon holds video sight (for daytime), thermal sight (for nighttime) and laser rangefinder capabilities. Land Warrior is years away from the troops in the field, but its replacement is already on the drawing boards. The designers of Future Warrior 2025 think they can make individual infantrymen as deadly as tanks and much more comfortable than today's foot soldiers. One day in the future, a Marine may be able to send cockroach-like robotic critters into a building to send back video images of the enemy. These creatures could also set off